

Measuring Polarization on Online Space Using Public Comments

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Abstract

Political polarization on social media has been extensively studied, yet most research focuses on single platforms, institutional political actors, or Western contexts. This study examines behavioral political polarization across Korean social media by comparing user interaction patterns across platform environments and political actor types. Using large-scale public comment datasets comprising 1,204,255 interactions collected from YouTube and Instagram communities centered around politicians and political influencers, this study constructs user-level measures of ideological tendency and cross-cutting exposure to evaluate how platform environment, producer type, and participation intensity relate to ideological segregation. Results show that YouTube communities exhibited stronger ideological segregation and lower cross-cutting engagement than Instagram communities across both actor types, suggesting that platform environments are associated with differing levels of behavioral polarization. Politician-centered communities, particularly on Instagram, demonstrated more ideologically mixed engagement than influencer-centered communities, indicating that institutional and non-institutional political actors organize audience interaction differently. Additionally, participation intensity revealed divergent patterns: highly active users in political YouTube communities showed disproportionately higher cross-cutting engagement than less active users, a pattern most consistent with oppositional monitoring behavior rather than deliberative openness. These findings contribute empirical evidence that behavioral polarization differs meaningfully across platform environments and political communities, extending polarization research beyond single-platform and Western-centered contexts.

Introduction

Political discussion has increasingly shifted onto social media platforms (Theocharis et al., 2023), where users engage with political actors not only through content consumption, but also through visible public interactions such as comments, replies, and discussions. Platforms such as Facebook, Twitter, and Instagram have become important

spaces for political communication, allowing politicians, influencers, and ordinary users to interact in networked online environments. As political discourse becomes more embedded within platform ecosystems, concerns surrounding online political polarization and ideological segregation have also grown (Kubin and Von Sikorski, 2021). Previous studies show that social media environments reinforce selective exposure, where users disproportionately engage with politically aligned content while avoiding opposing viewpoints (Stroud, 2010; Bail et al., 2018).

Among these platforms, YouTube has become an especially important source of political information for Korean audiences. A national survey of 2,538 respondents found that 46% access current events through YouTube, while 23% use the platform specifically to obtain political or societal information (Korean Statistical Information Service, 2025). Similarly, Reuters Institute for the Study of Journalism (2023) report shows that YouTube news consumption in South Korea is unusually high, with about 53% of respondents reporting that they obtain news through YouTube, compared to a global average of around 30%.

Existing studies on online polarization often focus on a single platform, a single political actor type, or Western political context (Falkenberg et al., 2025). Many studies also emphasized content-level, such as ideological framing or misinformation (Cinelli et al., 2021), rather than user-level interaction behavior. As a result, less is known about how users themselves interact across ideological boundaries in different platform environments, particularly within the context of Korean political social media. In addition, prior work rarely compares whether communities surrounding official politicians differ from those surrounding political YouTubers, despite their growing influence of nontraditional actors in shaping online discourse.

To address these gaps, this study examines political polarization across both social media platforms and content producer types in Korea. Specifically, this study compares interaction patterns on YouTube and Instagram across communities centered around politicians and political influencers. Using large-scale comment interaction datasets, this study constructs user-level measures of ideological engagement and cross-cutting exposure to evaluate how platform environments and content producer types shape patterns of online political interaction and ideological segregation. Specif-

ically, this study addresses three research questions: **RQ1** How do patterns of ideological segregation and cross-cutting exposure differ across YouTube and Instagram political communities? **RQ2** How do communities centered around politicians differ from those centered around political influencers in terms of behavioral polarization? and **RQ3** How does user participation intensity relate to ideological engagement and cross-cutting exposure across political communities?

To answer these questions, this study constructs user-level behavioral measures of ideological engagement using large-scale comment interaction datasets collected from Korean political communities across YouTube and Instagram. Users are further grouped by participation intensity to evaluate whether highly active users demonstrate different interaction patterns than less active users. By comparing platform environments, producer types, and user activity levels simultaneously, this study contributes empirical evidence toward understanding how platform structure and political community organization shape online political polarization.

This study makes four contributions. First, it provides a cross-platform comparison of behavioral political polarization across YouTube and Instagram within Korean political social media environments. Second, it compares ideological interaction patterns between communities centered around politicians and political influencers, extending polarization research beyond institutional political actors. Third, it integrates user-level behavioral measures of audience ideological tendency and cross-cutting exposure to capture multiple dimensions of online political interaction. Finally, this study introduces a large-scale dataset of Korean political social media comments and evaluates how participation intensity relates to ideological engagement patterns.

Related Works

Social media platforms reshaped political communication by enabling direct, networked interaction between political actors and the public, introducing new dynamics of ideological exposure and engagement (Papacharissi, 2010). Iyengar et al. (2019) and Kubin and Von Sikorski (2021), distinguish three mechanisms of online polarization: partisan network selection, algorithmic amplification of emotionally charged content, and platform specific design features that reward engagement over accuracy. Critically, scholarship distinguishes between ideological polarization, divergence in policy preference, and affective polarization, emotional hostility toward political outgroups, with the latter growing more rapidly (Wagner, 2021). Selective exposure compounds these effects: users disproportionately encounter political aligned contents, confirming existence beliefs while avoiding cross-cutting viewpoints (Stroud, 2010; Bail et al., 2018).

Boulianne et al. (2025), using cross-national survey data from four countries, find that ideological differences in political posting and content exposure are more noticeable across platforms than across national contexts, suggesting platform architecture may play a larger role than national context. YouTube is particularly consequential; its recommendation

algorithm and affordances for long-form opinionated content may concentrate ideologically aligned audiences, and patterns of audience migration and interaction suggest ideological clustering within political communities (Ribeiro et al., 2021). These findings suggest that platform-specific interaction environments may shape not only what users consume online, but also how users publicly engage with political content and opposing viewpoints. Baqir et al. (2023), suggests that visible engagement behaviors may capture different polarization dynamics than passive information consumption.

The emergence of political influencers as a distinct actor type introduces further complexity. Unlike elected politicians operating within institutional accountability structures, content creators may foster parasocial audience relationships that strengthen identification and engagement behaviors (Schramm et al., 2024). Loru et al. (2025), analyzing Twitter discussions across three global debates, finds that online polarization is structured more by users' ideological alignment with content producers than by the institutional category of those producers, suggesting influencer communities may polarize differently than politician-centered ones. Flamino et al. (2023) confirms this, showing that between 2016 and 2020 Twitter users increasingly clustered around ideologically aligned influencers, reducing cross-cutting encounters over time. Despite this evidence, no study has directly compared the community structures of politician- versus influencer-centered networks within the same platform environment, nor across platforms

South Korea offers an especially important test case. Approximately 50% of South Koreans regularly consume political news through YouTube—nearly double the global average—making it one of the most YouTube-dependent political media ecosystems in the world (Reuters Institute for the Study of Journalism, 2023; Korean Statistical Information Service, 2025). Tran et al. (2022, 2025) confirm that Korean political YouTube is characterized by distinct ideologically segregated user clusters, and that the platform's recommendation algorithm reinforces rather than bridges these divisions. Beyond platform dynamics, Korean polarization carries considerable geopolitical significance: Cheong and Haggard (2023) document that partisans in South Korea now extend to foreign policy orientations, with conservatives aligned toward the U.S.-Japan axis and progressives favoring engagement with China, a divide that maps directly onto the country's most consequential alliance relationships. Despite these stakes, empirical research on social media polarization remains mainly focused on Western, dominantly U.S. contexts (Falkenberg et al., 2025), leaving Korean platform dynamics comparatively understudied. This study addresses these gaps by comparing user-level ideological engagement and cross-cutting exposure across YouTube and Instagram, and across communities centered on politicians versus political influencers, contributing both to platform-specific polarization theory and to the growing literature on non-Western digital politics.

As researchers increasingly examine online political polarization beyond content characteristics alone, attention has also shifted toward behavioral measures that capture how

users interact across political communities. Prior work has increasingly emphasized behavioral measures of political polarization rather than relying solely on content-level analysis. User interaction patterns can reveal ideological segregation by examining how audiences distribute engagement across politically aligned and opposing communities. Existing work has operationalized polarization through measures of audience ideological consistency and cross-cutting exposure, capturing both ideological alignment and engagement across political boundaries (Robertson et al., 2018; Bakshy et al., 2015).

Methods

Dataset

In this study, the dataset consists of public comments collected from politicians and political content creators across both platforms. The list of current members of the assembly was collected from the government assembly website, and their YouTube channel and Instagram handle was collected manually. Members of the assembly were then labeled left, independent, and right, according to their parties. List of political content creators comes from NamuWiki, which serves as Wikipedia in Korea. The list came with labeled political ideology, which were manually verified while manually collecting their YouTube channel and Instagram handles.

After collecting YouTube channels and Instagram accounts, I collected user comments using YouTube Data v3 API, and Apify Instagram Comment Scraper.

The final dataset was organized into four groups based on platform environment and producer type: politician YouTube comments (N = 362,839), politician Instagram comments (N = 97,465), political YouTuber YouTube comments (N = 620,233), and political YouTuber Instagram comments (N = 123,718). The combined dataset contains 1,204,255 publicly observable comments spanning Korean political communities across both platforms.

The datasets are stored in JSON format and organized at the comment level. Each observation represents a single user interaction with a political post or video. Table 1 summarizes the representative schema structure across datasets. Example comment contents are omitted for space considerations but include comment text, timestamps, engagement metadata, and platform-specific identifiers.

It is common for comment authors to post a single comment and never appear again in political discussion spaces. Users with very few interactions provide insufficient behavioral data for reliable ideological inference, as the bias score is highly sensitive to individual interactions at low observation counts. A user with two comments on right-leaning channels, for instance, receives a score of 1.0 indistinguishable from a user with 200 such comments, despite representing a meaningless signal. To ensure that each user's score reflects a stable pattern of engagement rather than incidental activity, this study excludes users with fewer than 20 total interactions.

Example comment contents are omitted for space considerations but include comment text, timestamps, engagement metadata, and platform-specific identifiers.

Scope

This study focuses specifically on user commenting behavior rather than content sentiment or language modeling. Specifically, the analysis examines how users engage across ideological boundaries within Korean political communities on YouTube and Instagram by measuring interaction behavior through public comments. The objective is to evaluate how platform environments, content producer types, and participation intensity relate to ideological segregation and cross-cutting engagement patterns.

The analysis is limited to visible user interaction associated with politicians and political content creators. Political engagement is operationalized through user-level commenting behavior, where ideological tendencies are inferred from interaction patterns with labeled political actors rather than from self-reported political identity or manual interpretation of comment text. This allows large-scale behavioral comparison across platforms while maintaining a consistent measurement framework across different political communities.

Metrics

This study uses two primary metrics to measure political polarization: user bias score and cross-cutting exposures.

User Bias Score User ideological tendency is measured as:

$$\text{Bias} = \frac{R_i - L_i}{R_i + L_i}$$

Where R_i is the number of interactions with right-leaning channels/accounts and L_i is the number of interactions with left-leaning channels/accounts.

Scores closer to 1 indicate stronger right-leaning behavior, while scores closer to -1 indicate stronger left-leaning behavior.

This metric allows comparison of ideological consistency across users with different activity levels.

Cross-cutting Exposure Cross-cutting exposure is measured as:

$$\text{CrossCutting} = \frac{\text{Opposite}_i}{\text{Total}_i}$$

This metric captures the proportion of interactions users make with ideologically opposing content. Higher values indicate greater interaction with opposing viewpoints, while lower values indicate stronger ideological segregation.

To compare behavioral patterns across participation intensity, users were grouped into activity bands based on their total number of political interactions. Activity bands were defined as 20–49 interactions, 50–99 interactions, 100–299 interactions, 300–999 interactions, and 1000 or more interactions. Grouping users by participation intensity enables examination of whether highly active users demonstrate different patterns of ideological segregation and cross-cutting exposure compared to less active users.

Results

Distribution of User Ideology Across Platforms and Producer Types

Across all datasets, user ideology distributions generally exhibited polarized patterns, where many users clustered near

Platform	Producer	Source ID	Comment ID	User Field	Owner Field
YouTube	Politician	video_id	comment_id	author	channelName
YouTube	Political YouTuber	video_id	comment_id	author	channelName
Instagram	Politician	postUrl	commentUrl	ownerUsername	OwnerFullName
Instagram	Political YouTuber	postUrl	commentUrl	ownerUsername	OwnerFullName

Table 1: Representative schema structure across collected datasets. Each observation corresponds to a publicly observable user comment.

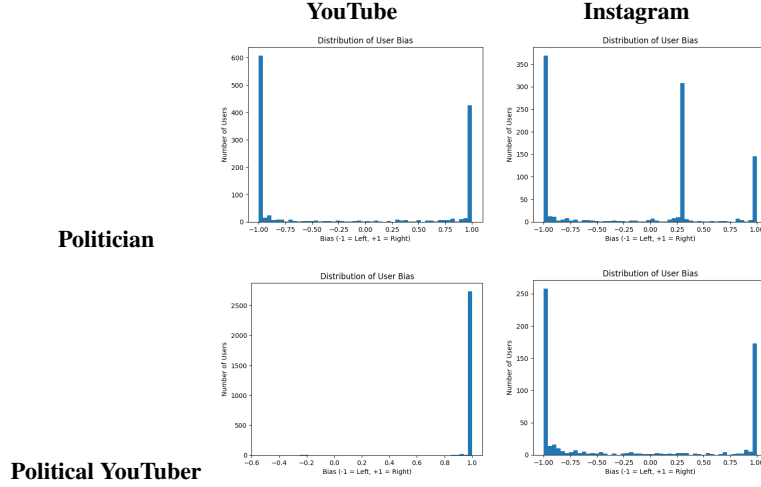


Figure 1: Distribution of user bias scores across platform and producer type.

the ideological extremes of -1 and 1 on the user bias score (Figure 1). This indicates that users frequently interacted with ideologically aligned political actors rather than engaging evenly across opposing sides.

As shown in Figure 1, politicians’ YouTube comment networks exhibited a strongly bimodal distribution, with large concentrations of users at both the left-leaning (-1) and right-leaning (+1) extremes. This suggests that political discussions on politicians’ YouTube channels were highly segregated by ideology.

Political YouTubers’ YouTube channels showed an even stronger concentration of users near the right-leaning extreme (+1), with comparatively fewer left-leaning users observed in the distribution. This indicates that interactions within political YouTuber communities were dominated by highly ideologically aligned users. One possible explanation is dataset coverage imbalance, as fewer active left-leaning users were observed and several left-leaning political YouTube communities restricted or disabled public comments.

Contrastingly, politicians’ Instagram accounts exhibited a more mixed ideological structure. As shown in Figure 1, while strong left- and right-leaning peaks remained visible, an additional concentration of users appeared around the moderate-right range (approximately 0.25), suggesting the presence of relatively more ideologically mixed engagement compared to YouTube environments.

Political YouTubers’ Instagram accounts also displayed polarized patterns, though the distribution appeared more dispersed than political YouTubers’ YouTube channels.

Compared to YouTube, Instagram interactions showed a broader spread of users across intermediate ideological values rather than clustering primarily at ideological extremes.

Overall, Figure 1 suggests that YouTube communities demonstrated stronger ideological concentration, while Instagram communities exhibited comparatively broader behavioral distributions.

Distribution of Cross-Cutting Exposure

User-level cross-cutting exposure distributions showed that most users engaged very little with ideologically opposing actors across all four datasets (Figure 2). In each platform and producer type, the majority of users exhibited cross-cutting exposure values close to zero, indicating limited interaction with opposing viewpoints. The distributions suggest that observable political engagement remained concentrated within ideologically aligned communities across both platforms.

As shown in Figure 2, the politicians’ Instagram dataset exhibited a visible secondary concentration around moderate cross-cutting levels. This pattern suggests that some users interacted with both ideological sides rather than remaining fully segregated, indicating relatively greater ideological overlap compared to the other communities examined.

Overall, the findings suggest that Instagram demonstrated higher observed cross-cutting exposure levels than YouTube, although users across all platforms and producer types still exhibited generally low levels of cross-cutting engagement. These findings indicate that while platform environment

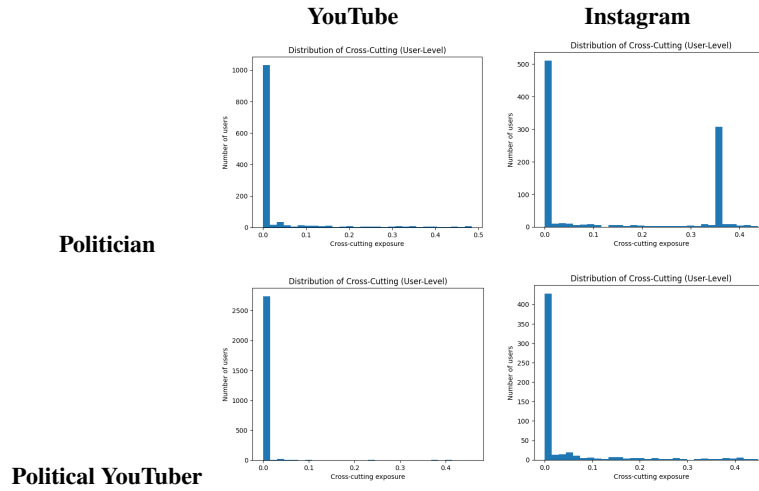


Figure 2: Distribution of cross-cutting exposure across platform and producer type.

may influence opportunities for interaction across ideological boundaries, observable engagement behavior remained predominantly ideologically concentrated.

Cross Cutting Exposure by User Activity Level

Cross-cutting exposure was further examined across user activity levels to evaluate whether highly active users engaged differently from less active users (Figure 3). User activity was grouped into interaction bands based on total activity volume.

As shown in Figure 3, politicians' YouTube channels exhibited consistently low cross-cutting exposure across all activity bands, with only modest variation between less active and highly active users. Users with 20–49 interactions exhibited an average cross-cutting exposure of 0.028 (95% CI $\pm$ 0.005), while users with 50–99 interactions showed a slightly higher average exposure of 0.043 (95% CI $\pm$ 0.019). Users in the 100–299 activity range demonstrated a similar average exposure of 0.040 (95% CI $\pm$ 0.037).

Although the 300–999 and 1000+ activity bands appeared to show lower cross-cutting exposure values, these categories contained only a single user each and therefore should be interpreted cautiously. Overall, Figure 3 suggests that even highly active users on politicians' YouTube channels engaged minimally with ideologically opposing actors.

In contrast, Figure 3 shows that political YouTubers' YouTube channels exhibited extremely low cross-cutting exposure among low-activity users but notable increase in cross-cutting exposure among highly active users. Users in the 20–49 interaction band demonstrated an average cross-cutting exposure of 0.001 (95% CI $\pm$ 0.0004), while users in the 50–99 interaction band showed nearly zero cross-cutting exposure overall. Users with 100–299 interactions, however, exhibited substantially higher average cross-cutting exposure of 0.134 (95% CI $\pm$ 0.077).

The 300–999 activity band showed an even higher observed cross-cutting exposure value of 0.376; however, this category contained only a single user and therefore cannot

be generalized. No users were observed in the 1000+ activity band. Collectively, these findings suggest that while most users within political YouTuber communities interacted almost exclusively with ideologically aligned content, a small subset of highly active users demonstrated considerably greater cross-cutting engagement.

For politicians' Instagram accounts, Figure 3 indicates generally higher cross-cutting exposure than observed on YouTube platforms across most activity bands. Users in the 20–49 interaction band exhibited average cross-cutting exposure of 0.098 (95% CI $\pm$ 0.013), while users in the 50–99 and 100–299 interaction bands demonstrated higher average exposures of 0.224 (95% CI $\pm$ 0.019) and 0.235 (95% CI $\pm$ 0.043), respectively.

The 300–999 activity range showed moderate average cross-cutting exposure of 0.182, although this estimate was associated with considerable uncertainty due to relatively small sample size ($n = 6$). The 1000+ activity band contained only a single user and therefore should not be interpreted cautiously. Overall, these findings suggest that Instagram interactions surrounding politicians were associated with greater ideological overlap and cross-cutting engagement than comparable YouTube environments, particularly among moderately active users.

Political YouTubers' Instagram accounts exhibited relatively stable cross-cutting exposure across lower and moderate activity bands. Users in the 20–49 interaction band demonstrated average cross-cutting exposure of 0.040 (95% CI $\pm$ 0.009), while users in the 50–99 and 100–299 interaction bands exhibited similar exposure levels of 0.042 (95% CI $\pm$ 0.022) and 0.043 (95% CI $\pm$ 0.044), respectively.

Users in the 300–999 activity range demonstrated higher average cross-cutting exposure of 0.113; however, this estimate was based on only three users and therefore should be interpreted cautiously. No users were observed in the 1000+ activity band. Figure 3 suggests that ideological engagement patterns within political YouTubers' Instagram communities remained comparatively stable across most participation lev-

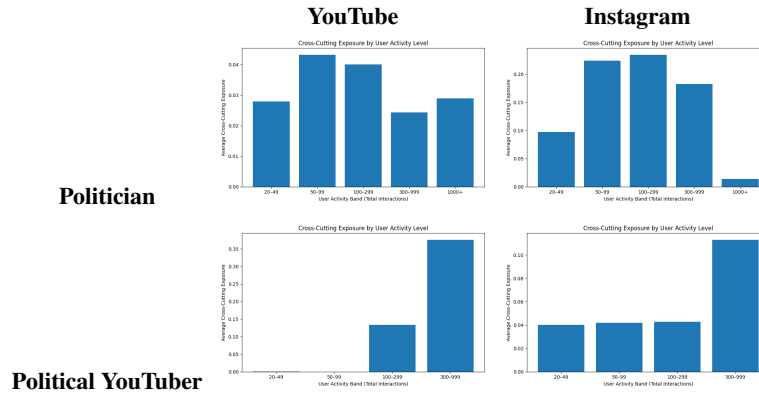


Figure 3: Cross-cutting exposure across activity bands by platform and producer type.

els, with generally modest levels of cross-cutting exposure.

Discussion

This study examined behavioral political polarization across Korean social media environments by comparing interaction patterns across platforms, political actor types, and user participation intensity. Across all four datasets, users interacted disproportionately with ideologically aligned communities, producing strongly polarized audience distributions and limited cross-cutting engagement. Ideological segregation is a structural feature of Korean political social media, observable at the user interaction level across both platforms and both producer types.

The findings provide clear evidence that the platform environment shapes behavioral polarization. Across both politicians and political YouTubers, YouTube communities exhibited stronger ideological segregation and lower cross-cutting exposure than Instagram communities. YouTube audiences clustered near ideological extremes, while Instagram communities showed broader ideological distributions and meaningfully higher cross-cutting engagement — particularly in politician-centered accounts, where a visible concentration of users appeared in the moderate-right range alongside the expected partisan extremes. These differences align with Boulianne et al. (2025) argument that platform architecture exerts stronger influence on political behavior than national context, and with Ribeiro et al. (2021) account of YouTube’s recommendation system concentrating on ideologically aligned audiences.

Producer types also structured interaction patterns, particularly on YouTube. Political YouTuber communities demonstrated more concentrated ideological engagement than politician-centered communities, especially the strong right-skew in YouTubers’ YouTube channels, though this must be read cautiously given that several left-leaning channels had their comment sections disabled, limiting the observable left-leaning user base. The more significant novel finding is that politician-centered Instagram communities showed the most ideologically mixed engagement of all four datasets. Politicians, as institutional actors with broader constituency obligations, appear to attract more ideologically diverse commenting audiences than influencers, who

frequently cultivate creator-centered communities organized around shared interests, identities, and recurring engagement patterns (Riedl et al., 2023; Flamino et al., 2023). This supports treating producer type as a meaningful analytical variable in polarization research rather than collapsing all political content creators into a single category.

The participation intensity findings reveal the most theoretically interesting pattern. On politician-centered YouTube channels, cross-cutting exposure remained flat across all activity bands, even the most active users engaged minimally with ideologically opposing content, indicating stable and deeply entrenched segregation regardless of engagement level. On political YouTubers’ YouTube channels, however, cross-cutting exposure was near zero among low-activity users but jumped to 0.134 among users in the 100–299 interaction band. This non-linear pattern reflects: a small subset of highly engaged users actively seeking out opposing-side content for purposes of criticism, counter-mobilization, or surveillance rather than genuine ideological openness. This interpretation is consistent with Baqir et al. (2023) observation that visible interaction behavior captures distinct dynamics from passive consumption. cross-cutting exposure, in other words, should not be equated with deliberative openness; hostile engagement also registers as cross-cutting in behavioral measures.

Taken together, platform environment produced the strongest and most consistent variation in behavioral polarization, while producer type and participation intensity shaped how users engaged across ideological boundaries within those environments. These findings extend prior scholarship by demonstrating, through large-scale behavioral data, that online political polarization in South Korea is not uniform across digital spaces, its degree and structure vary meaningfully depending on where users engage and with whom.

Several limitations should be acknowledged. This study measures only publicly observable commenting behavior and therefore cannot capture passive information consumption behaviors such as content viewing, scrolling behavior, subscription activity, or recommendation-driven exposure pathways. Consequently, cross-cutting exposure as operationalized in this study should be interpreted as observ-

able interaction across ideological boundaries rather than total ideological exposure, as users may encounter opposing viewpoints without visibly engaging with them (Baqir et al., 2023). Ideological orientation is inferred behaviorally rather than measured through self-reported identity, and the right-skew in YouTubers' YouTube data may partly reflect the comment-disabling constraint rather than community structure alone. These findings are specific to Korean political social media and may not generalize directly to other contexts. Future work should incorporate longitudinal analysis, passive consumption measures, and sentiment-level examination of cross-cutting exposures to build on these findings.

Conclusion

This study compared behavioral political polarization across YouTube and Instagram communities centered around Korean politicians and political influencers. Using large-scale public comment data, it constructed user-level measures of ideological tendency and cross-cutting exposure across four distinct community types. Three findings stand out. First, YouTube consistently produced greater ideological segregation and lower cross-cutting engagement than Instagram across both producer types, pointing to platform architecture as a primary structural determinant of behavioral polarization. Second, politician-centered communities, particularly on Instagram, showed more ideologically mixed engagement than influencer-centered communities, suggesting that institutional and non-institutional political actors organize audience interaction differently. Third, participation intensity revealed divergent patterns depending on platform and producer type: segregation was stable across activity levels on politicians' YouTube, while highly active users in YouTuber communities showed disproportionately higher cross-cutting engagement — most plausibly reflecting oppositional monitoring rather than deliberative openness.

These findings contribute to a growing literature on platform-specific and non-Western digital politics by demonstrating that behavioral polarization is shaped not only by user preferences but by the structural environments in which political engagement occurs. As platform design decisions increasingly determine the architecture of political interaction, understanding how those structures shape ideological segregation is essential for evaluating their democratic consequences.

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